
SOFTWARE REQUIREMENTS SPECIFICATION

for
Cafeteria Management System

Version 1.0

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Submitted to:

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1 Introduction

1.1 Purpose

The primary objective of this document is to outline the requirements of the cafeteria management system. This document provides a detailed description of both functional and non-functional requirements as proposed by the client. The purpose of this project is to streamline cafeteria operations, including order processing, inventory management, and payment handling, by replacing manual processes with an efficient, computer-based system. It aims to enhance user experience for cafeteria staff and customers while generating useful reports for management. This project uses Entity-Relationship (ER) diagrams and Unified Modeling Language (UML) diagrams to describe hardware and software interface requirements.

1.2 Document Conventions

- **Main Title:** Times New Roman, Bold, Size 14
- **Sub Title:** Times New Roman, Bold, Size 12
- **Body Text:** Times New Roman, Size 12

1.3 Scope of Development Project

The Cafeteria Management System is designed to transform traditional cafeteria operations into an internet-based application, allowing users to place orders, track inventory, and manage payments seamlessly. It targets cafeteria staff, customers (students/staff), and administrators. The project is developed using Java for its performance and scalability.

1.4 Definitions, Acronyms, and Abbreviations

- **JAVA:** Platform-independent programming language
- **SQL:** Structured Query Language
- **ER:** Entity Relationship
- **UML:** Unified Modeling Language
- **IDE:** Integrated Development Environment
- **SRS:** Software Requirement Specification
- **POS:** Point of Sale
- **IEEE:** Institute of Electrical and Electronics Engineers

1.5 References

Books:

- *Software Requirements and Specifications* by Michael Jackson
- *Software Engineering: A Practitioner's Approach* by Roger S. Pressman

Websites:

- <http://www.slideshare.net/>
- <http://ebookily.net/doc/srs-cafeteria-management-system>

2 Overall Descriptions

2.1 Product Perspective

The Cafeteria Management System supports two primary user types: cafeteria staff (administrators) and customers (students/staff). The system offers search functionality for menu items, order placement, and inventory tracking.

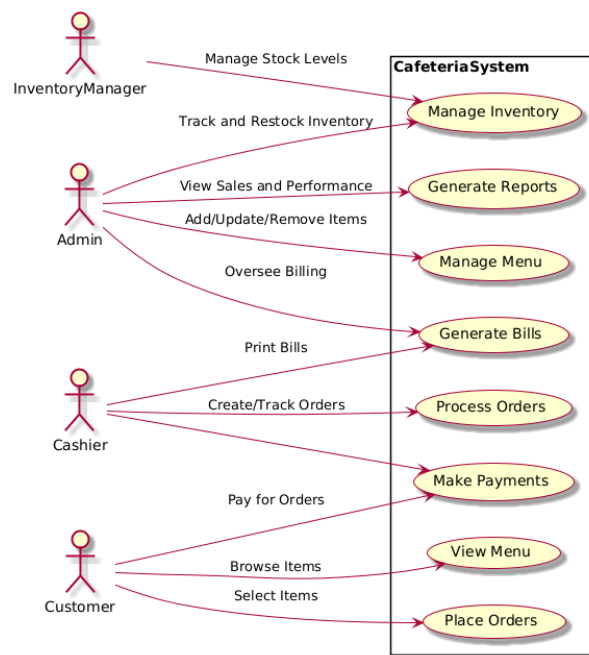


Figure 2.1: Use Case Diagram

2.2 Product Function

The system provides real-time information on menu availability, order status, and inventory levels. Its primary goal is to reduce manual effort in order processing, payment collection, and stock management.

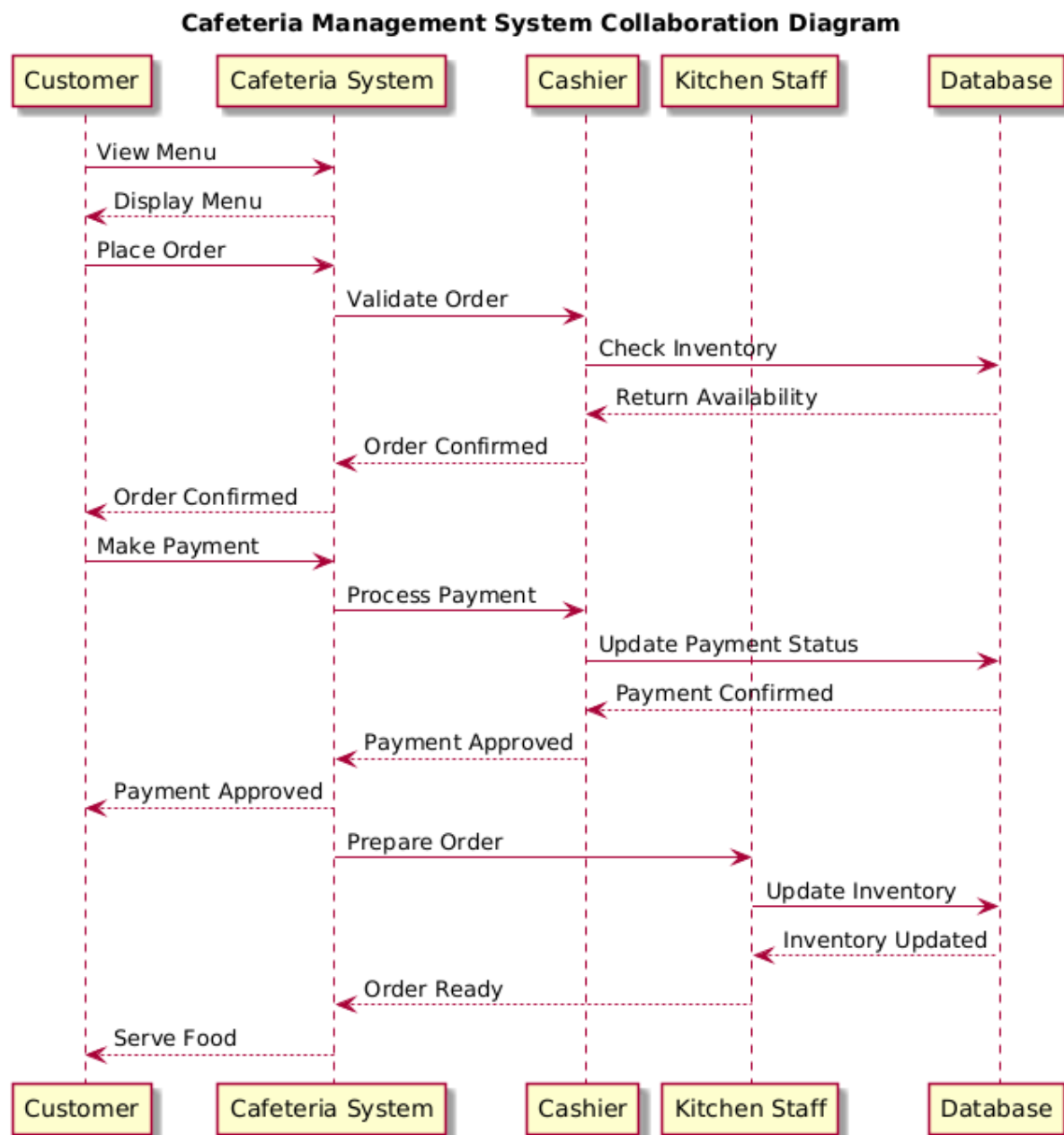


Figure 2.2: Collaboration Diagram

2.3 User Classes and Characteristics

The system caters to:

- **Administrator (Cafeteria Staff):** Manage the system, process orders, update inventory.
- **Customers (Students/Staff):** Browse the menu, place orders, and track order status.

2.4 Operating Environment

The system operates on Windows OS, accessible via modern web browsers like Edge, Chrome, and Firefox. Minimum hardware requirements include a 40 GB hard disk and a 15" monitor.

2.5 Assumptions and Dependencies

Assumptions:

- The system is error-free and user-friendly.
- Accurate and timely database updates.

3 System Features

- **Security:** Role-based access control and password protection.
- **Monitoring:** Alerts for low stock or payment issues.

4 External Interface Requirements

4.1 GUI

The system offers a user-friendly interface for both administrators and customers with search and report generation features.

5 Other Non-functional Requirements

5.1 Performance Requirement

The system must handle multiple simultaneous orders, process payments quickly, and manage large datasets.

5.2 Safety Requirement

Regular database backups are mandatory to prevent data loss due to crashes or power failures.

5.3 Security Requirement

- Uses a secure database.
- Implements role-based access control.

5.4 Glossary

- **Administrator:** User with full system privileges
- **POS:** Point of Sale system for payments

5.5 Class Diagram

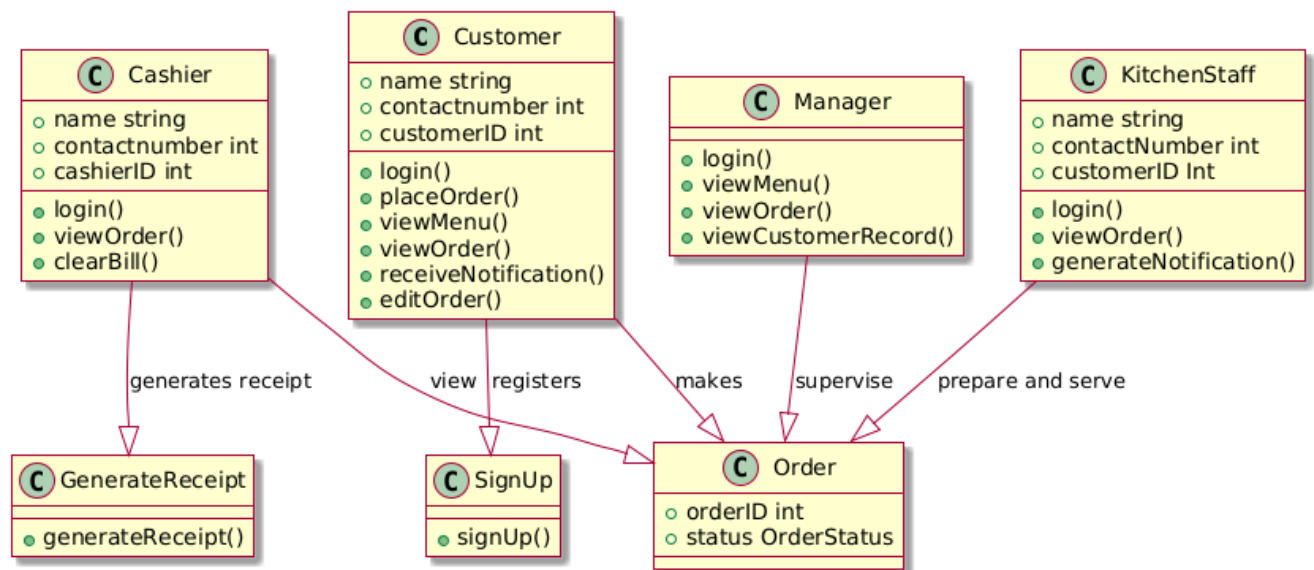


Figure 5.5: Class Diagram